

# Daniel Felipe Vargas Ulloa

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Double-degree last year engineering student at École Centrale Nantes and University of the Andes with industry experience in data analysis, machine learning, and software development. Seeking a 6-month final year internship from April to September 2027.

## EDUCATION

**École Centrale de Nantes** Nantes, France  
*Diplôme d'Ingénieur (Master's level Engineering Degree)* 2025 – 2027

- Specialization in Computer Science for Information Systems

**University of the Andes** Bogotá, Colombia  
*Bachelor's in Systems and Computer Engineering* 2021 – 2027  
*Minor in Computational Mathematics* GPA: 4.4 / 5.0

- National Recognition among Top-Performing Students (Colombian Ministry of Education, 2024)

## EXPERIENCE

**Climate Risk Assessment of Colombia's Hydroelectric Infrastructure** May 2026 – Sept 2026  
*University Jorge Tadeo Lozano* Bogotá, Colombia

- Processed and analyzed NASA satellite observations and climate data to assess the vulnerability of Colombia's hydroelectric infrastructure to hydroclimatic stress.
- Developed quantitative indices to assess reservoir vulnerability and quantify the impact of El Niño-related drought on Colombia's hydroelectric system.

**Clinical Anomaly Detection System for Pediatric Intensive Care Units Project** Jan 2025 – Jul 2025  
*Central Military Hospital of Bogotá* Bogotá, Colombia

- Developed a real-time predictive monitoring system for pediatric vital signs, reducing the false positive rate by 38.7% and improving early detection of clinical deterioration in pediatric patients.
- Built end-to-end pipelines for data ingestion, signal processing, and detection of early clinical deterioration to support early intervention strategies.

**Undergraduate Teaching Assistant, Software Development** Jan 2024 – Jul 2024  
*University of the Andes* Bogotá, Colombia

- Co-delivered a production oriented course on designing, implementing, and deploying full-stack applications using React, Spring microservices, and REST APIs.
- Supported development teams in applying Agile methodologies, version control using Git, test-driven development, and containerized deployment using Docker.

## SKILLS

**Programming Languages:** Python, SQL, Java, C/C++, Go

**Data & Machine Learning:** Signal Processing, Predictive Modeling, Anomaly Detection, Data Pipelines, Power BI

**Backend & Data Systems:** Spring Boot, Flask, REST APIs, Docker, PostgreSQL, MongoDB, Linux, Git, Google Cloud

## LANGUAGES

**Spanish** – Native · **English** – C2 (Fluent) · **French** – B2 (Advanced)